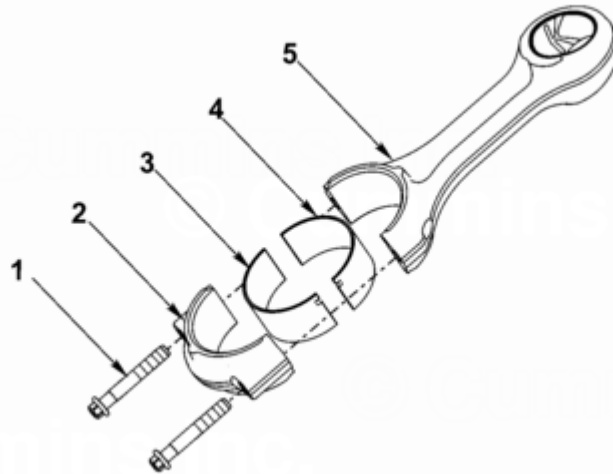


Trainings ▾

Exploded View



01r00249

Connecting Rod Bearings Exploded View

1. Capscrew - quantity 2
2. Connecting rod cap - quantity 1
3. Lower connecting rod bearing shell - quantity 1
4. Upper connecting rod bearing shell - quantity 1
5. Connecting rod - quantity 1

Select Service Tools

Recommended Cummins® Service Tools

- Engine barring tool, Part Number 4919092
- Connecting rod guide pin, Part Number 5298634
- Lubriplate™ 105 multi-purpose lubricant, Part Number 3163087, or equivalent

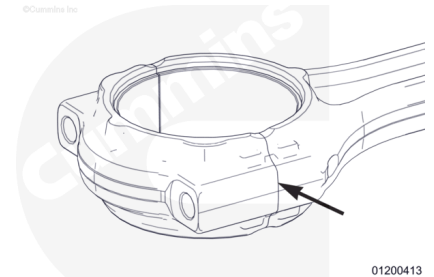
Additional Service Items

- Rubber hammer
- Awl
- Fine grit crocus cloth

- Clean engine oil.

General Information

The connecting rod for this engine uses a fracture-split connection between the connecting rod cap and connecting rod. Connecting rods with a fracture-split surface **must** be treated with caution. The two pieces of the connecting rod can **not** be rubbed together. The mating surfaces will be damaged. Be careful **not** to drop either piece of the connecting rod. Fracture-split connecting rods **must only** be handled if the two pieces of the connecting rod are tightened to the correct specification or completely separated.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

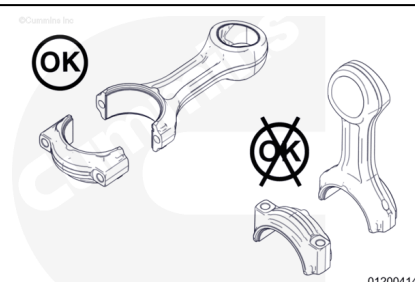
Rotating the engine with the connecting rod caps loose or removed can cause damage to the piston cooling nozzles. Always remove the piston cooling nozzles prior to removing the connecting rod and piston assembly.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the lubricating oil system. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/1016/1016-007-037.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-089.html)
- Remove the piston cooling nozzles. Refer to Procedure 001-046 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-046.html)

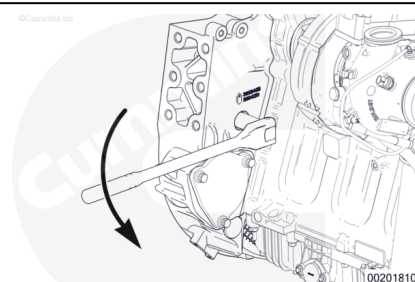
Remove

⚠ CAUTION ⚠

To prevent damage to the fracture-split connecting rod, do not set the connecting rod or rod cap on the fracture-split connection. Polishing and damage to the mating surface can result.

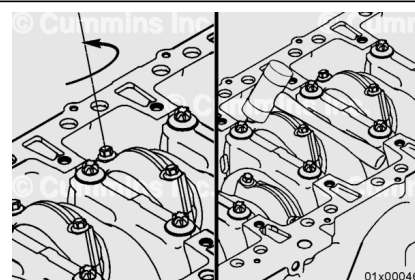


Rotate the crankshaft so two of the connecting rods are at bottom dead center. Use engine barring tool, Part Number 4919092.



⚠ CAUTION ⚠

Do not damage the fracture-split surface on the connecting rod or connecting rod cap while the connecting rod cap is removed. If the fracture-split surface is damaged, the connecting rod and connecting rod cap must be replaced to reduce the possibility of engine damage. Incorrect assembly can damage the connecting rod.



⚠ CAUTION ⚠

Verify the piston cooling nozzles have been removed before removing the connecting rod cap. Damage to the nozzles can occur if the nozzles are not removed.

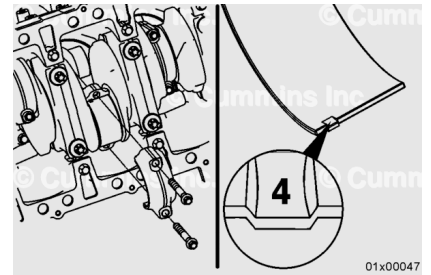
Loosen the connecting rod capscrews.

Do **not** remove the capscrews at this time.

Hit the connecting rod capscrews with a rubber hammer to loosen the connecting rod cap.

⚠ CAUTION ⚠

Minimize the number of joint interactions between the connecting rod and connecting rod cap. If the fracture-split mating surface becomes damaged or worn, the connecting rod cap will not fit as designed to the connecting rod. Engine damage can occur.



Remove the capscrews and connecting rod cap. Note the corresponding connecting rod and connecting rod cap as well as the orientation of the connecting rod cap.

Be careful **not** to damage the fracture-split connection.

Do **not** set the connecting rod cap down on its fracture-split face.

Both the connecting rod cap and connecting rod are serialized on one face to aid in proper orientation.

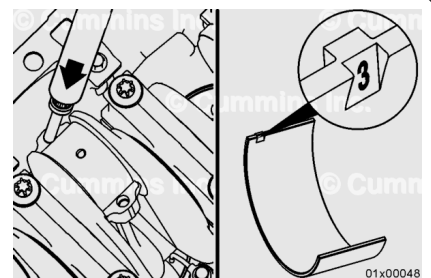
The lower connecting rod bearing is denoted by LOWER or LWR, for the connecting rod cap, stamped into the back of the bearing.

Remove the lower bearing from the lower connecting rod cap. Use an awl to mark the cylinder number on the flat surface of the bearing tang.

Use connecting rod guide pin, Part Number 5298634, to push the rod up far enough to allow the upper bearing to be removed.

Do **not** push against the fracture-split face of the rod.

Use caution to avoid scratching or other damage to the journal with the connecting rod, capscrew, or connecting rod guide pin.



The upper connecting rod bearing is denoted by UPPER or UPR, for the connecting rod, stamped into the back of the bearing. The upper connecting rod bearing can also be identified by the oil port in the bearing.

Remove the bearing. Mark the cylinder number on the flat surface of the bearing tang.

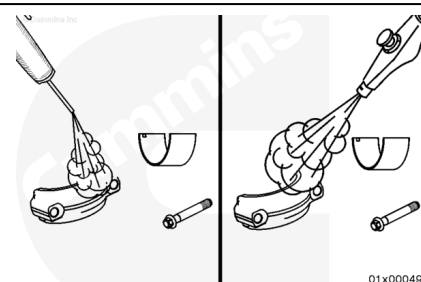
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Steam clean the parts. Dry with compressed air.

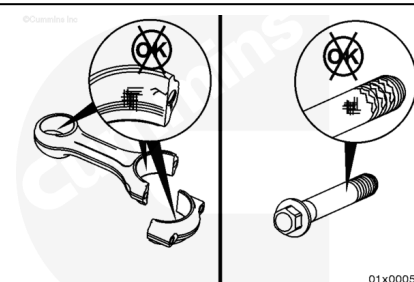
Inspect the connecting rod caps, connecting rod bearing saddles, and capscrews for nicks, cracks, burrs, scratches, and fretting.

Inspect the fracture surface for damage and signs of polishing.

Polished spots on either the connecting rod or connecting rod cap mating surfaces indicate fretting and indicate the connecting rod **must** be replaced.

Dark areas on the face indicate carbonized oil deposits. Connecting rods with dark areas are reusable if fretting is **not** present.

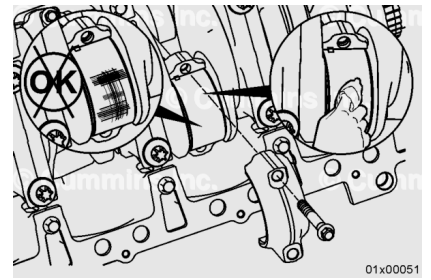
Polishing can be observed on the thrust face of the connecting rod or connecting rod cap, which is normal. Use the following procedure for inspection criteria. Refer to Procedure 001-007 in Section 1. (</qs3/pubsys2/xml/en/procedures/1016/1016-001-007.html>)



Inspect the connecting rod journal of the crankshaft for scratches or nicks.

Use a fine grit crocus cloth to remove the nicks and scratches.

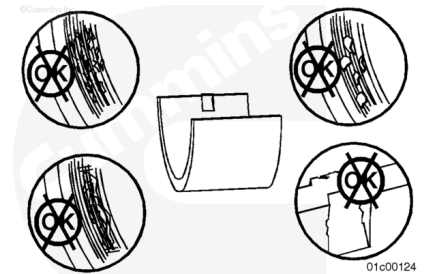
Clean off any dust or debris released while using the crocus cloth.



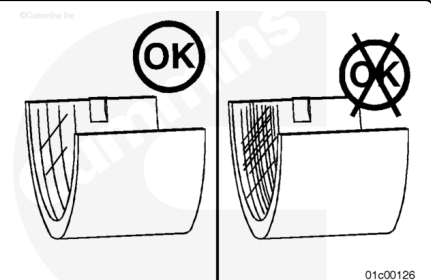
Inspect the bearings for damage.

Replace any bearings with the following damage:

- Pitting
- Flaking
- Corrosion
- Lock tang damage
- Scratches (deep enough to be felt with a fingernail).



Normal bearing wear produces a smooth finish that will wear into the lining. Refer to Service Bulletin, Parts Reuse Guidelines, Bulletin 3810303 (</qs3/pubsys2/xml/en/bulletin/3810303.html>).



Refer to Service Bulletin, Analysis and Prevention of Bearing Failures, Bulletin 3810387 (</qs3/pubsys2/xml/en/bulletin/3810387.html>), for more detailed information of bearing damage.

Inspect the bearing seating surface for nicks or burrs.

If nicks or burrs can **not** be removed with a fine crocus cloth, the bearings **must** be replaced.

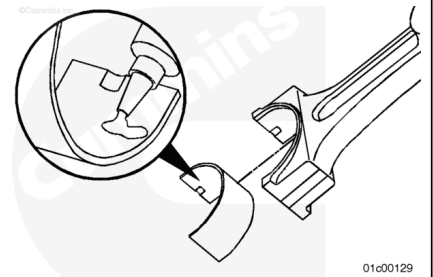
Clean off any dust or debris released while using the crocus cloth.



Install

⚠ CAUTION ⚠

The connecting rod and bearing mating surfaces must be clean and dry when the bearings are installed. Used bearings must be installed in their original location or engine damage can occur.

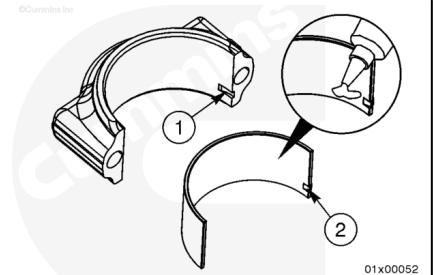


Use clean Lubriplate™ 105 multi-purpose lubricant, Part Number 3163087, or equivalent, to lubricate the crankshaft journal mating surface of the upper bearing.

Install the upper bearing into the connecting rod with the tang of the bearing in the slot of the connecting rod.

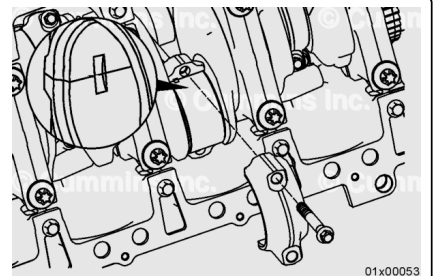
Install the bearing into the connecting rod cap with the tang (2) of the bearing in the slot (1) of the connecting rod cap.

Use clean Lubriplate™ 105 multi-purpose lubricant, Part Number 3163087, or equivalent, to lubricate the bearing to the crankshaft journal mating surface.



⚠ CAUTION ⚠

Do not damage the fracture-split surface on the connecting rod or connecting rod cap while the connecting rod cap is removed. If the fracture-split surface is damaged, the connecting rod and connecting rod cap must be replaced to reduce the possibility of engine damage. Incorrect assembly can damage the connecting rod.



⚠ CAUTION ⚠

Minimize the number of joint interactions between the connecting rod and connecting rod cap. If the fracture-split mating surface becomes damaged or worn, the connecting rod cap will not fit as designed to the connecting rod. Engine damage can occur.

Verify the connecting rod cap is installed **only** on the connecting rod from which the connecting rod cap is removed and the connecting rod cap is properly oriented during installation.

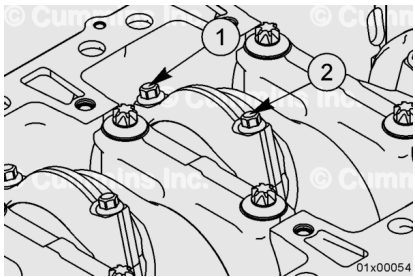
Both the connecting rod cap and connecting rod are serialized on one face to aid in proper orientation.

Install the connecting rod cap onto the connecting rod.

Lubricate the threads and the underside of the heads of the connecting rod capscrews. Use clean engine oil.

Thread the capscrews into the connecting rod. Hand tighten.

Tighten the capscrews.



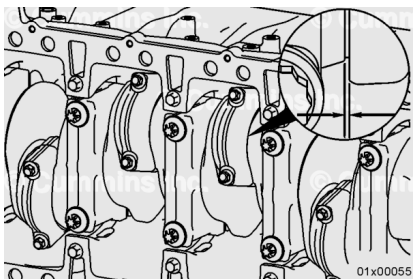
Torque Value:

1. 91 N•m [67 ft-lb]
2. Rotate 60 degrees.

Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-054.html)

Measure the connecting rod side clearance.

Connecting Rod Side Clearance		
mm		in
0.10	MIN	0.004
0.35	MAX	0.014



The connecting rod **must** move freely from side to side on the crankshaft journal. If the connecting rod does **not** move freely, remove the connecting rod cap and verify the bearings are the correct size. Check for dirt or damage on the crankshaft and the bearings.

Repeat the above steps to install the remaining bearings and connecting rod caps.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ CAUTION ⚠

The lubricating oil system must be primed before operating the engine after any internal engine repairs to reduce the possibility of internal component damage.

- Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-046.html)
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-089.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html)
- Fill the lubricating oil system. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/1016/1016-007-037.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.